

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO4: Evaluate model-based reinforcement learning approaches for prediction, planning, and policy optimization. (BL3,BL4,BL5)

Assessment Tool Weightage: 40%

Dr G. A. SENTHIL

QUESTIONS: 25

Total Marks: 50

Annexure A

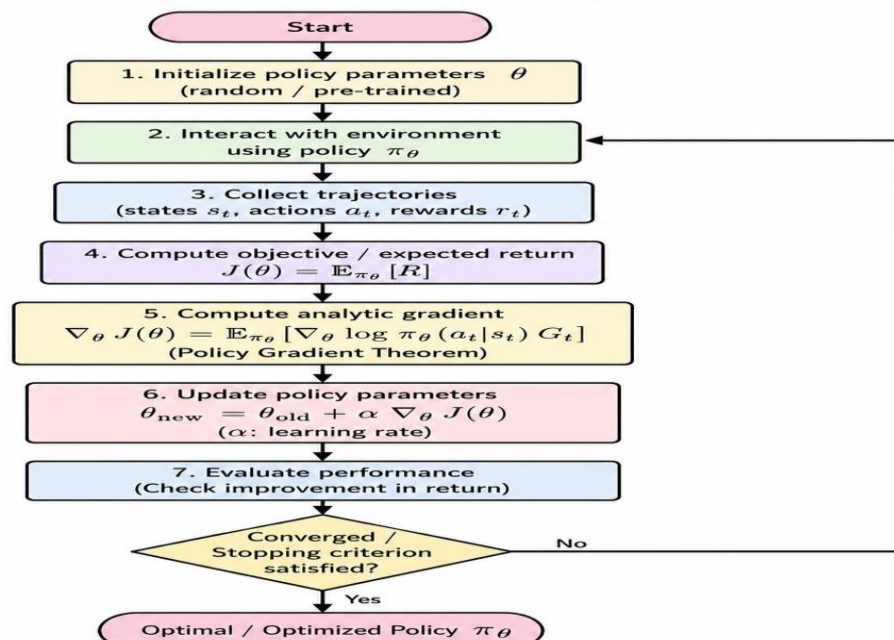
Design and Develop Model

DURATION: 4 HRS (2 Sessions)

Analytic Gradient Computation

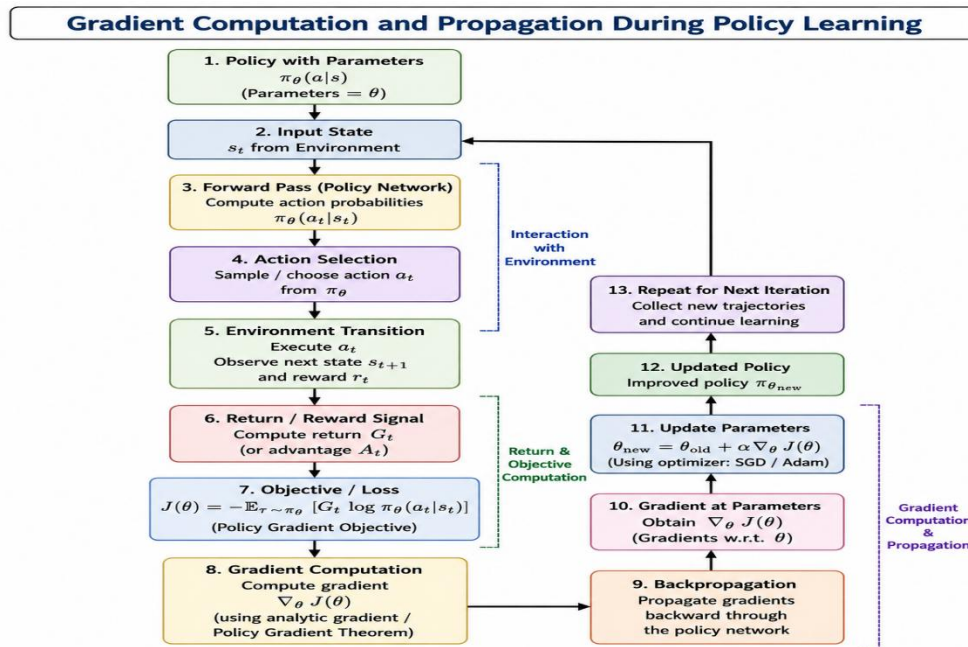
6. Draw a flowchart explaining the steps involved in analytic gradient computation for policy optimization

Analytic Gradient Computation for Policy Optimization



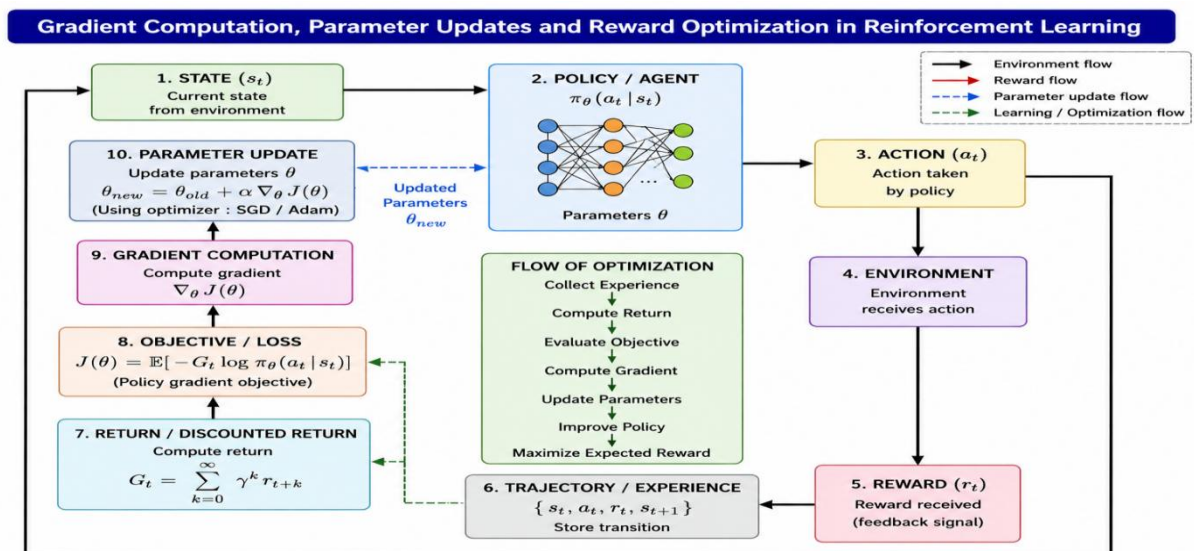
(5 Marks)

7. Design a concept map showing how gradients are computed and propagated during policy learning.



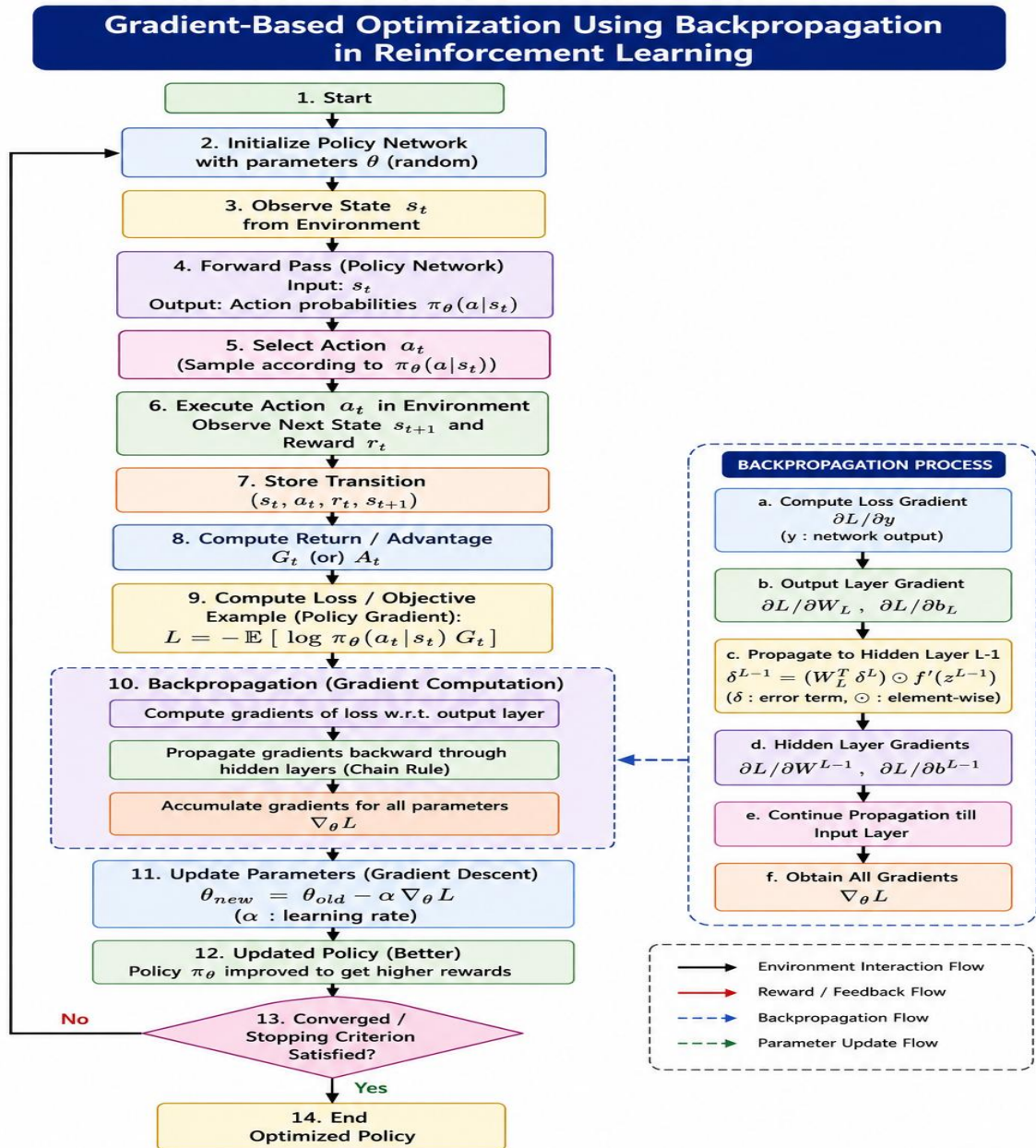
(5 Marks)

8. Draw a block diagram illustrating gradient computation, parameter updates, and reward optimization in Reinforcement Learning.



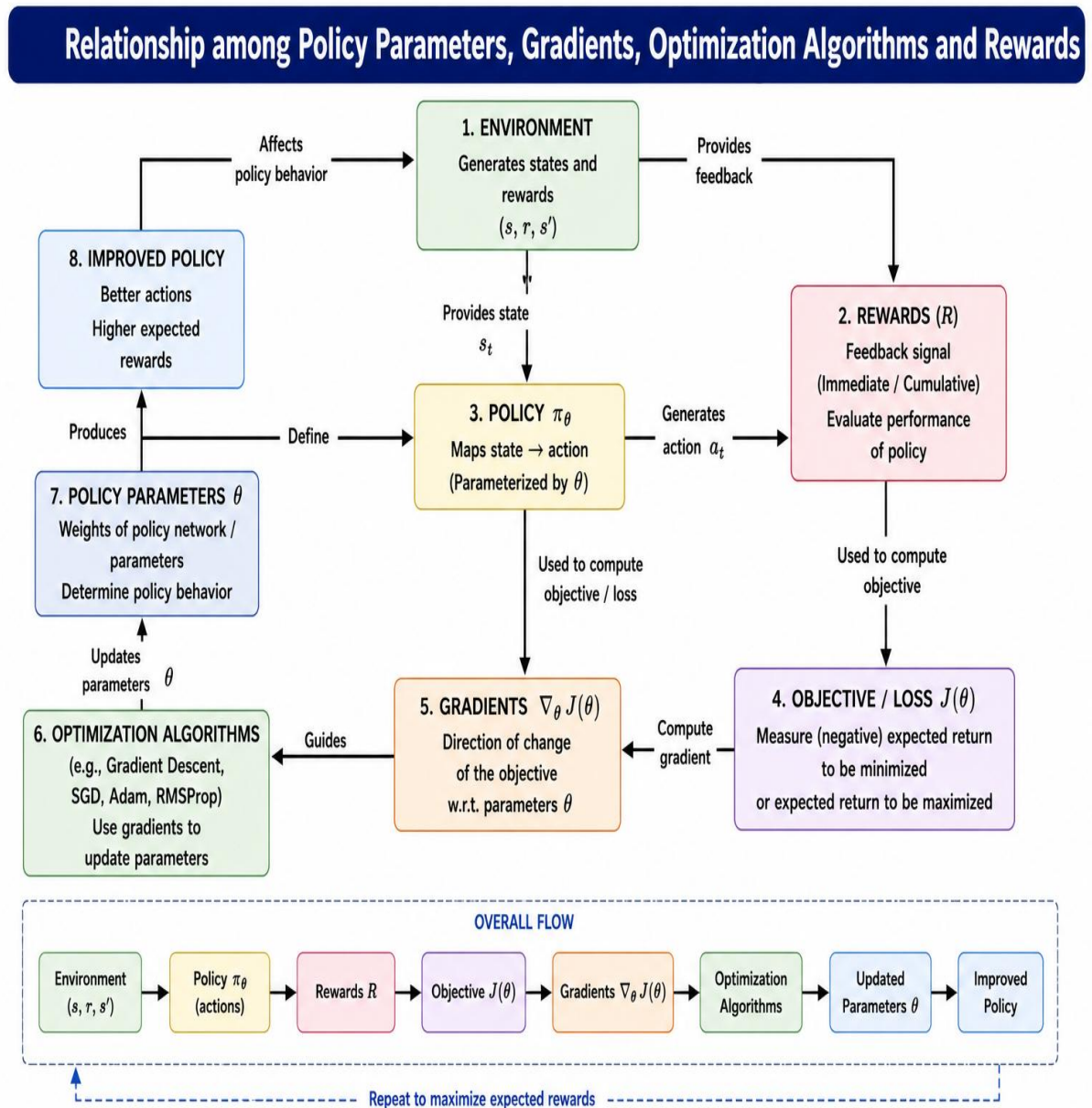
(5 Marks)

9. Prepare a flowchart explaining gradient-based optimization using backpropagation in Reinforcement Learning.



(5 Marks)

10. Draw a concept map showing the relationship among policy parameters, gradients, optimization algorithms, and rewards.



(5 Marks)

Code of Conduct:

I, K.Mounith(192425090) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Mounith

RUBRIS FOR CALCULATION

Criteria	Weightage	Level – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness